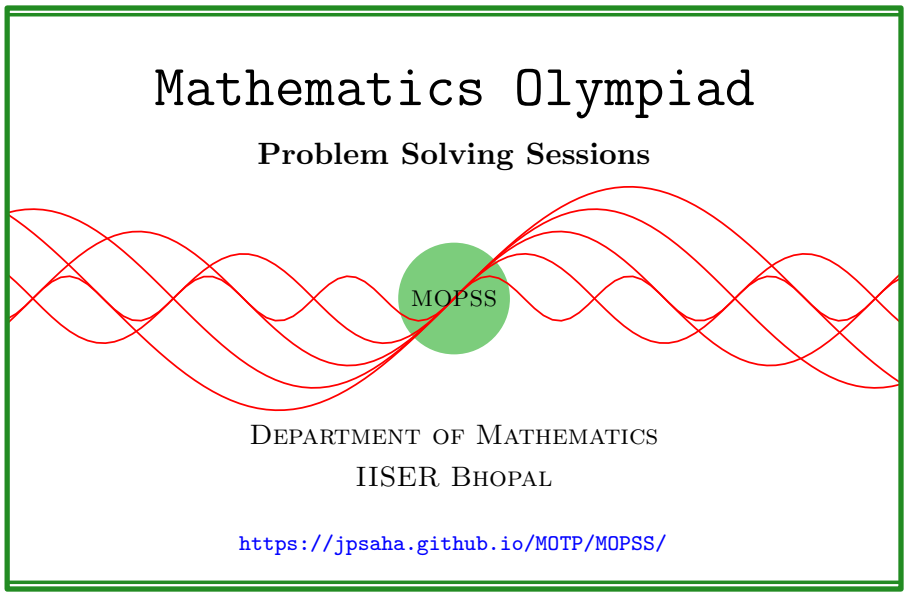
Integer divisibility

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Integer divisibility

Exercise 1. If $P(x)$ is a polynomial with integer coefficients and a, b are integers, then $P(a) - P(b)$ is a multiple of $a - b$.

Walkthrough —

(a) Show that it suffices to prove it for monomials.

Solution 1. ■

Exercise 2 (RMO 2015c P3,). Let $P(x)$ be a nonconstant polynomial whose coefficients are positive integers. If $P(n)$ divides $P(P(n) - 2015)$ for all natural numbers n, then prove that $P(-2015) = 0$.

Summary — In absolute value, a higher degree polynomial dominates a smaller degree polynomial at arguments which are large enough in absolute value.

Solution 2. Let $Q(x), R(x)$ be polynomials with rational coefficients such that

$$P(P(x) - 2015) = P(x)Q(x) + R(x)$$

and $R(x) = 0$ or $\deg R(x) < \deg P(x)$. Note that $P(n)$ is positive for all integer $n \geq 1$ since the coefficients of $P(x)$ are positive integers. By the given condition, it follows that $P(n)$ divides $R(n)$ for any integer $n \geq 1$.

Claim — Let $f(x), g(x)$ be two polynomials with real coefficients. Suppose $f(x)$ is a nonconstant polynomial with a positive leading coefficient, and

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> $\deg g(x) < \deg f(x)$. Then there exists an integer $n_0 \geq 1$ such that
>
> $$f(n) > g(n)$$
>
> for any $n \geq n_0$.

Proof of the Claim. Note that it suffices to proves the Claim if $f(x)$ is a monomial, that is, a power of x. Indeed, write $f(x) = a_d x^d + a_{d-1}x^{d-1} + \cdots + a_0$ with $a_0, \ldots, a_d \in \mathbb{R}$ and d denoting the degree of f. Also write $g(x) = b_e x^e + b_{e-1}x^{e-1} + \cdots + b_0$ with $b_0, \ldots, b_e \in \mathbb{R}$ and e denoting the degree of g. Noting that $a_d > 0$, it follows that for a positive integer n, the inequality

$$a_d n^d + a_{d-1}n^{d-1} + \cdots + a_0 > b_e n^e + b_{e-1}n^{e-1} + \cdots + b_0$$

holds if

$$a_d n^d > \frac{a_{d-1}}{a_d}n^{d-1} + \cdots + \frac{a_0}{a_d} + \frac{b_e}{a_d}n^e + \frac{b_{e-1}}{a_d}n^{e-1} + \cdots + \frac{b_0}{a_d}$$

is satisfied, which can be concluded provided the Claim is known in the case when f is a monomial.

Let us assume that f is a monomial. Write $f(x) = x^d$ where d denotes the degree of f, and write $g(x) = b_e x^e + b_{e-1}x^{e-1} + \cdots + b_0$ with $b_0, \ldots, b_e \in \mathbb{R}$ and e denoting the degree of g. For any integer n, note that

$$\begin{aligned} f(n) - g(n) &= \left(\frac{1}{e+1}n^d - b_e n^e\right) \\ &\quad + \left(\frac{1}{e+1}n^d - b_{e-1}n^{e-1}\right) \\ &\quad + \cdots + \left(\frac{1}{e+1}n^d - b_0\right) \\ &\geq \left(\frac{1}{e+1}n^d - |b_e|n^e\right) \\ &\quad + \left(\frac{1}{e+1}n^d - |b_{e-1}|n^{e-1}\right) \\ &\quad + \cdots + \left(\frac{1}{e+1}n^d - |b_0|\right). \end{aligned}$$

Since $d \geq e$, it follows that there exists an integer $n_0 \geq 1$ such that

$$\frac{1}{e+1}n^d - |b_e|n^e \frac{1}{e+1}n^d - |b_{e-1}|n^{e-1}, \ldots, \frac{1}{e+1}n^d - |b_0|$$

are positive for any $n \geq n_0$. This proves the Claim. □

By the above Claim, it follows that $R(x)$ is the zero polynomial. This implies that

$$P(P(x) - 2015) = P(x)Q(x).$$

Since $P(x)$ is a nonconstant polynomial, it has a root z in $\mathbb{C}$. Substituting $x = z$ yields

$$P(-2015) = 0.$$

■

Here is an alternate solution.

Solution 3. Since $P(x)$ is a polynomial with integer coefficients, it follows that for any integers a, b, the difference $P(a) - P(b)$ is divisible by $a - b$. This shows that

$$P(P(n) - 2015) - P(-2015)$$

is divisible by $P(n)$ for all natural numbers n. Since $P(n)$ divides $P(P(n)-2015)$ for all natural numbers n, it follows that $P(n)$ divides $P(-2015)$ for all natural numbers n. Let d denote the degree of $P(x)$, and a_d denote the leading coefficient of $P(x)$. Since $P(x)$ has positive integer coefficients, it follows that

$$P(n) \geq a_d n^d$$

holds for any natural number n. Taking $k = |P(-2015)|$, we have that $P(k+1)$ divides k. Since $P(k+1) \geq a_d(k+1)^d > k$ and k is a nonnegative integer, we conclude that $k = 0$, which implies that $P(-2015) = 0$. ■

Exercise 3 (Flanders Mathematical Olympiad 2025 P3,). Consider the polynomial $P(x) = 1 + x + x^2 + ... + x^{2023} + x^{2024}$. What is the remainder polynomial when $P(x^{2025})$ is divided by $P(x)$?

> **Walkthrough** — Show that $P(x^{2025}) - 1$ is divisible by $P(x)$.

Solution 4. Note that $P(x^{2025}) - P(1)$ is divisible by $x^{2025} - 1$ and hence by $P(x)$. Therefore, the remainder polynomial when $P(x^{2025})$ is divided by $P(x)$ is equal to $P(1) = 2025$. ■

Example 4. Let $P(x)$ be a polynomial with integer coefficients such that $P(0), P(1)$ are odd. Show that $P(x)$ does not have any integer root.

Solution 5. If $P(x)$ admits an integer root α, then α divides $P(0)$ and $\alpha - 1$ divides $P(1)$, which shows that $\alpha, \alpha - 1$ are odd, which is impossible. ■

Exercise 5 (IOQM 2023 P12,). Let $P(x) = x^3 + ax^2 + bx + c$ be a polynomial where a, b, c are integers and c is odd. Let p_i be the value of $P(x)$ at $x = i$. Given that $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, find the value of $p_2 + 2p_1 - 3p_0$.

Summary — Using parity conditions, show that $p_1 + p_2 + p_3 \neq 0$. This gives $p_1 = p_2 = p_3$. This gives a system of equations in a and b, which can be solved to find a and b.

Walkthrough —

(a) Note that if $p_1 + p_2 + p_3 \neq 0$, then using the condition that p_1, p_2, p_3 are real, it follows that $p_1 = p_2 = p_3$.

(b) Note that

$$\begin{aligned} p_1 &= 1 + a + b + c, \\ p_2 &= 8 + 4a + 2b + c, \\ p_3 &= 27 + 9a + 3b + c, \end{aligned}$$

which gives that

$$p_1 + p_2 + p_3 = 36 + 14a + 6b + 3c.$$

Since c is an odd integer, it follows that[a] $p_1 + p_2 + p_3 \neq 0$. Consequently, $p_1 = p_2 = p_3$.

(c) This gives

$$3a + b = -7, 5a + b = -19,$$

and hence, $a = -6, b = 11$.

(d) This gives

$$\begin{aligned} p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\ &= 18. \end{aligned}$$

[a] Since c is odd, it follows that p_2 is odd. Note that $p_1 - p_3$ is even (Why? Does it follow that $f(1) - f(3)$ is divisible by 2 for any polynomial $f(x)$ with integer coefficients?), and hence, $p_1 + p_2 + p_3$ is odd.

Solution 6. Since $P(x)$ is a polynomial with integer coefficients, it follows that $p_1 - p_3$ is divisible by 2. Since c is odd, it follows that p_2 is odd. This shows that $p_1 + p_2 + p_3$ is odd, and hence, $p_1 + p_2 + p_3 \neq 0$. Using $p_1^3 + p_2^3 + p_3^3 = 3p_1p_2p_3$, we obtain

$$(p_1 + p_2 + p_3)(p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1) = 0.$$

This gives

$$p_1^2 + p_2^2 + p_3^2 - p_1p_2 - p_2p_3 - p_3p_1 = 0,$$

oe equivalently,

$$(p_1 - p_2)^2 + (p_2 - p_3)^2 + (p_3 - p_1)^2 = 0,$$

which implies that $p_1 = p_2 = p_3$. This shows that

$$1 + a + b + c = 8 + 4a + 2b + c = 27 + 9a + 3b + c,$$

which yields

$$3a + b = -7, 5a + b = -19.$$

This gives $a = -6, b = 11$. It follows that

$$\begin{aligned} p_2 + 2p_1 - 3p_0 &= 10 + 6a + 4b + 3c - 3c \\ &= 10 - 36 + 44 \\ &= 18. \end{aligned}$$

■

Exercise 6 (RMO 2016g P8,). At some integer points a polynomial with integer coefficients take values $1, 2$ and 3. Prove that there exist not more than one integer at which the polynomial is equal to 5.

Solution 7. Denote the polynomial by $P(x)$. On the contrary, let us assume that there are at least two distinct integers where $P(x)$ takes the value 5.

Let a, b, c be integers such that

$$P(a) = 1, P(b) = 2, P(c) = 3.$$

Note that $a - b$ divides $P(a) - P(b)$, $b - c$ divides $P(b) - P(c)$. It follows that $a - b = \pm 1, b - c = \pm 1$. Since a, b are of opposite parity, and so are the integers b, c, we obtain that a, c are of the same parity. Noting that $c - a$ divides $P(c) - P(a) = 2$, it follows that $c - a = \pm 2$. Combining this with $a - b = \pm 1, b - c = \pm 1$, we get $a - b = b - c = 1$ or $a - b = b - c = -1$.

This shows that $P(b - 1) = 1, P(b) = 2, P(b + 1) = 3$ holds or $P(b + 1) = 1, P(b) = 2, P(b - 1) = 3$ holds. Note that in the first case, the polynomial $R(x) := P(x - b)$ takes the values $1, 2, 3$ at the integers $-1, 0, 1$ respectively. In the second case, the polynomial $S(x) = P(-x + b)$ takes the values $1, 2, 3$ at the integers $-1, 0, 1$ respectively. This proves that there is a polynomial $Q(x)$ with integer coefficients which takes the values $1, 2, 3$ at $-1, 0, 1$ respectively.

From the hypothesis, it follows that there are distinct integers i, j such that $Q(i) = Q(j) = 5$. Note that $i - 1$ divides $Q(i) - Q(1) = 2$, i divides $Q(i) - Q(0) = 3$, $i + 1$ divides $Q(i) - Q(-1) = 4$. Since i divdes 3, we obtain $i = \pm 1, \pm 3$. Using $Q(-1) = 1, Q(1) = 3$, we get $i \neq -1, i \neq 1$. This gives $i = \pm 3$. Noting that $i - 1$ divides 2, we obtain $i \neq -3$, and hence $i = 3$. Similarly, it follows that $j = 3$. ■

Example 7. Let $P(x)$ be a polynomial with integer coefficients such that $P(20), P(25)$ are of absolute value equal to 1. Show that $P(x)$ does not vanish at any integer.

Walkthrough —

(a) If $P(x)$ vanishes at an integer α, then $\alpha - 20$ divides $P(20)$ and $\alpha - 25$ divides $P(25)$.

Solution 8. On the contrary, let us assume that $P(x)$ vanishes at an integer α. Note that $\alpha - 20$ divides 1, and so does $\alpha - 25$. This shows that $\alpha - 20, \alpha - 25$ are of absolute value equal to 1. Applying triangle inequality, we obtain

$$5 \leq |\alpha - 20| + |\alpha - 25| \leq 2,$$

which is impossible. ■

Exercise 8 (USAMO 1974 P1,). Let a, b, and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Walkthrough —

(a) Show that each of the integers $a - b, b - c, c - a$ is a multiple of the remaining two integers.

(b) Prove that this implies that a, b, c are equal.

Solution 9. Note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) \mid c - a \mid P(c) - P(a) = a - b.$$

Consequently, the integers $a - b, b - c, c - a$ are of the same absolute value. Denote their absolute value by k. Note that their sum is zero. However, the sum is equal to mk, for some $m \in \{\pm 1, \pm 3\}$. Hence, k is equal to zero. This yields that $a = b = c$. ■

Here is a more general result.

Example 9. Let $P(x)$ be a polynomial with integer coefficients, and let n be an odd positive integer. Suppose that $x_1, x_2, \ldots, x_n$ is a sequence of integers such that $x_2 = P(x_1), x_3 = P(x_2), \ldots, x_n = P(x_{n-1})$, and $x_1 = P(x_n)$. Prove that all the x_i's are equal.

Walkthrough — Show that

$$a_1 - a_2 \mid a_2 - a_3 \mid a_3 - a_4 \mid \cdots \mid a_n - a_1 \mid a_1 - a_2.$$

Note that sum of these differences is an odd multiple of their absolute value.

Exercise 10 (Tournament of Towns Spring 2014, Senior A Level P4, by G. K. Zhukov)**.** In the plane, the points with integer coordinates (x, y) satisfying $0 \leq y \leq 10$ are marked. Consider a polynomial of degree 20 with integer coefficients. Determine the maximum possible number of marked points which can lie on its graph.

Walkthrough —

(a)

Solution 10. Note that the polynomial

$$(x-1)(x-2)(x-3)\dots(x-20)$$

of degree 20 has integer roots. Let us prove that the graph of no polynomial of degree 20 with integer coefficients passes through more than 20 marked points.

Claim — Let $P(x)$ be a polynomial of degree 20 with integer coefficients. No more than 20 marked points lie on the graph of $P(x)$.

Proof of the Claim. On the contrary, let us assume that there are integers $x_1 < x_2 < \cdots < x_{21}$ such that

$$0 \leq P(x_i) \leq 10$$

holds for all $1 \leq i \leq 21$. For any integer $1 \leq i \leq 10$, the inequality $x_{21} - x_i \geq 11$ holds, and using that $x_{21} - x_i$ divides the integer $P(x_{21}) - P(x_i)$, which lies in $[-10, 10]$, it follows that $P(x_{21}) = P(x_i)$. Similarly, for any integer $12 \leq i \leq 21$, it follows that $P(x_1) = P(x_i)$. This shows that $P(x_i) = P(x_1)$ for any integer $i \in \{1, 2, \dots, 10\} \cup \{12, 13, \dots, 21\}$. Since $P(x)$ is a polynomial of degree 20, it follows that

$$P(x) - P(x_1) = c(x-x_1)(x-x_2)\dots(x-x_{10})(x-x_{12})(x-x_{13})\dots(x-x_{21})$$

holds for some nonzero integer a. This yields that

$$|P(x_{11}) - P(x_1)| \geq (10!)^2,$$

which is impossible. □

This proves that the maximum possible number of marked points which can lie on the graph of a polynomial of degree 20 with integer coefficients is equal to 20. ■

Exercise 11 (RMO 2025b P6,). Let $p(x)$ be a nonconstant polynomial with integer coefficients, and let $n \geq 2$ be an integer such that no term of the sequence

$$p(0), p(p(0)), p(p(p(0))), \ldots$$

is divisible by n. Show that there exist integers a, b such that $0 \leq a < b \leq n-1$ and n divides $p(b) - p(a)$.

Walkthrough —

(a) Apply the pigeonhole principle to show that some two terms of the given sequence are congruent modulo n.

(b) Let i and j be positive integers such that $i < j$ and the i-th and j-th terms of the sequence are congruent modulo n.

(c) Reduce the arguments of these two terms modulo n to obtain integers a and b such that $0 \leq a, b \leq n-1$ and n divides $p(b) - p(a)$.

(d) Show that $a \neq b$. It could be useful to assume that the integer i as above is the smallest with this property.

Solution 11. Define the sequence

$$p^{(0)}(0), p^{(1)}(0), p^{(2)}(0), p^{(3)}(0), \ldots$$

by

$$p^{(n)}(0) \coloneqq \begin{cases} 0 & \text{if } n = 0, \\ p(p^{(n-1)}(0)) & \text{if } n \geq 1. \end{cases}$$

By the pigeonhole principle, some two among the following $n+1$ integers

$$p^{(1)}(0), p^{(2)}(0), \ldots, p^{(n+1)}(0)$$

are congruent modulo n. Let i denote the smallest positive integer such that there exists a positive integer $j > i$ with

$$p^{(i)}(0) \equiv p^{(j)}(0) \pmod{n}.$$

Let $0 \leq a \leq n-1$ and $0 \leq b \leq n-1$ be integers such that

$$a \equiv p^{(i-1)}(0) \pmod{n}, \quad b \equiv p^{(j-1)}(0) \pmod{n}.$$

Note that n divides $p(b) - p(a)$. We claim that $a \neq b$. Suppose, for the sake of contradiction, that $a = b$. It follows that

$$p^{(i-1)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Using $j - 1 > i - 1$ and the minimality of i, we obtain $i = 1$. This implies that

$$p^{(0)}(0) \equiv p^{(j-1)}(0) \pmod{n}.$$

Since $p^{(0)}(0) = 0$ and $j - 1 > 0$, we have that n divides $p^{(j-1)}(0)$, contradicting the given condition. This implies that $a \neq b$.

Reordering if necessary, we may assume that $a < b$. Consequently, there exist integers $0 \leq a < b \leq n - 1$ such that n divides $p(b) - p(a)$. ■

Lemma 1

Let P be a polynomial with integer coefficients. Suppose a is an integer and k is a positive integer such that $P^k(a) = a$, where P^k denotes the k-fold composite map from $\mathbb{Z} \to \mathbb{Z}$. Show that $P^2(a) = a$.

Proof. Let ℓ denote the smallest positive integer such that $P^\ell(a) = a$. If $\ell = 1$ or $\ell = 2$, then we are done. Henceforth, we assume that $\ell \geq 3$.

Note that

$$P(a) - a \mid P^2(a) - P(a) \mid \cdots \mid P^\ell(a) - P^{\ell-1}(a) = a - P^{\ell-1}(a) \mid P(a) - a.$$

Since $a - P^{\ell-1}(a)$ is nonzero, it follows that the above differences are nonzero. Consequently, for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = \pm(P^i(a) - P^{i-1}(a)).$$

If $P^{i+1}(a) = P^{i-1}(a)$ holds for some $1 \leq i \leq \ell$, then applying $P^{\ell-i+1}$ to both sides, we obtain $P^2(a) = a$, which contradicts the assumption that $\ell \geq 3$. It follows that for any $1 \leq i \leq \ell$,

$$P^{i+1}(a) - P^i(a) = P(a) - a$$

holds, which implies that

$$\sum_{i=0}^{\ell-1} (P^{i+1}(a) - P^i(a)) = \ell(P(a) - a).$$

This gives $P(a) = a$, which contradicts the assumption that $\ell \geq 3$. This completes the proof. □

Exercise 12 (IMO 2006 P5, by Dan Schwarz, Romania)**.** Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients, and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\ldots P(P(x))\ldots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Walkthrough —

(a) Does the above lemma help?

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Solution 12. By the above lemma, it reduces to considering the case $Q(x) = P^2(x)$.

Suppose Q has more than n fixed points. Since P is not linear, it follows that P cannot have n fixed points, and hence not all the fixed points of Q are fixed points of P. Let b be a non-fixed point of P, and $Q(b) = b$. Suppose a be a fixed point of Q, other than b.

Let us first consider the case that $P(a) \neq a$. Note that

$$P(b) - a \mid P(a) - b \mid P(b) - a$$

holds, and

$$a - b \mid P(a) - P(b) \mid a - b$$

holds too. This yields that

$$|P(a) - b| = |P(b) - a|, \quad |P(a) - P(b)| = |a - b|.$$

If

$$P(a) - b = a - P(b), \text{ and } P(a) - P(b) = a - b$$

hold, then b would be a fixed point of P. It follows that at least one of

$$P(a) - b = -(a - P(b)), P(a) - P(b) = -(a - b)$$

holds. Consequently, we obtain

$$P(a) + a = P(b) + b.$$

Next, let us consider the case that $P(a) = a$. Note that

$$P(b) - a \mid b - a \mid P(b) - a.$$

Since b is not a fixed point for P, it follows that

$$P(b) - a = a - b,$$

which yields

$$P(a) + a = P(b) + b.$$

This proves that all the roots of $Q(x) = x$ are the roots of $P(x) + x = P(b) + b$. Since $P(x)$ has degree $n > 1$, it follows that the polynomial $P(x) + x - P(b) - b$ is of degree n, and it has more than n roots, which is impossible.

Hence, there are at most n integers t such that $Q(t) = t$ holds. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)